Outlook enters Android Wear

Microsoft's Outlook email app has been launched for Android Wear Devices making it easier to reply with pre-set messages. **http://dgit.in/OutDev**

China's realistic robot

China's realistic robot Jia Jia revealed which can have conversations with humans and can even pose for pictures. **http://dgit.in/JiaJiaR**

when asked whether the obsolescence of Moore's law plays a part. "We are increasingly less reliant on the implicit benefits provided by moving down to a new process node. Purpose built processors and perhaps even more importantly, elegant marriage of software on top of the underlying silicon is going to be a huge source of improvements going forward, both from a technological and economic perspective."

Movidius is a small chip startup that you've never heard of. It's an European company that currently produces specialised chips, known as Vision Processing Units (VPUs). The chips are meant for application in areas like Augmented Reality, Virtual Reality, etc. Movidius' Myriad 2 chip is running on the recently announced DJI Phantom 4 drone and its Myriad 1 chip was used in Google's first Project Tango device. Dashwood explains that while Movidius is a decade old company, it turned its attention to VPUs in 2009, with Google's first Project Tango phone being the first time its chips were implemented. The Myriad 1 and Myriad 2 are high performance, but low power chips, meant specifically for Computer Vision.

The failure of Moore's law has led companies to look at new methods of extrapolating trends at the silicon level. The Myriad 2 chip is one such promising example. VPUs like the Myriad 2 don't sound like much, but like GPUs, they come out of a specific need for computer vision. Computer Vision is a branch of computing that deals with processing and understanding real world elements and images. It's the technology behind the myriad of augmented reality apps that you see today, like Blippar. It's also important in intelligent drones and robots that can navigate around real world obstacles and interpret real world objects by themselves.

"GPUs are actually quite a good analogy. In the early 90's, people realised that 3D gaming and visualisation was going to be hugely important for both commercial and consumer purposes, but existing architectures were not well suited to the types of computation required for rich graphics. In a similar vein, we are

A Vision Processing Unit or VPU

now keenly aware of the value of computer vision, but much of the existing hardware and software approaches aren't optimised for such tasks," says Dashwood.

VPUs are to Computer Vision, what GPUs are to gaming and graphics.

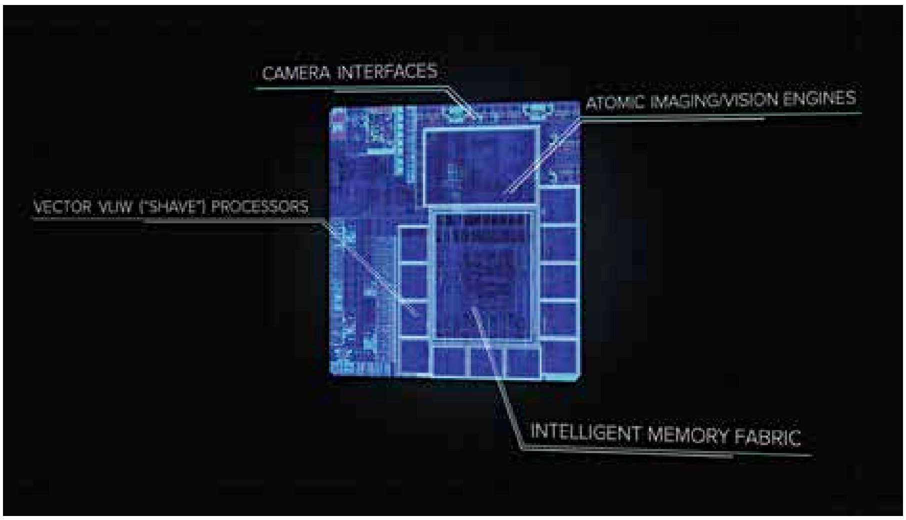
Obstacles to overcome

Using a specialised chip for a particular purpose is easier said than done though. Gamers usually use complex liquid cooling techniques in order to meet the heat requirements that GPUs come with. But while GPUs were originally meant for PCs, where there was enough space to implements such cooling methods, VPUs do not enjoy that luxury. They are meant for drones, smartphones and other smaller devices. They are meant to be an integral part of the mobile environment that the world is rapidly progressing towards.

According to Dashwood, that problem is already solved. "The Myriad 2 is developed from the ground up to run in a low power envelope, and at temperatures low enough that they can be embedded on wearable devices." The Myriad 2 can process millions of pixels, while consuming less than one watt of power. This is significantly lower than the power consumed by smartphone processors today and is necessary for a chip that is supposed to run alongside those processors. In essence, while the multi-core processor on your phone will be responsible for the fast boot-up of an augmented reality app, the VPU will be responsible for what that app does.

Heat isn't the only hurdle though. The problem with implementing specialised chips is that it's harder to program for them. Dashwood explains that the Myriad 2 is aimed at device manufacturers who are already competent in this realm. Programmability of the chip shouldn't be confused with end-user applications running on the Android OS, he says.

The computing industry is no stranger to specialised logic though. Intel's newest chips have special programming, meant for videos and other tasks, MediaTek's Helio chips come with the CorePilot algorithm to improve performance,



VPUs help improve the camera quality of smartphones